

Does the Depth Finding Survive at Eight Layers?

A CLS decomposition and attention-suppression ablation of the `layers8` arm

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Abstract

The 16-layer MARINA model does almost no work in its first ten cross-attention blocks: suppressing them costs 0.0072 cosine, while suppressing the last block alone costs 0.0368. That raised an obvious question — is the depth simply unused, so that a shallower model would behave identically? This note answers it by repeating the same two measurements on two independently trained 8-layer checkpoints. The answer is that depth is *not* wasted so much as *redistributed*. The number of blocks that actually carry computation is 5–6 in both models, and the work those final blocks do is quantitatively almost identical across depths — suppressing the last k blocks costs the same cosine at 8 layers as at 16, for every $k \leq 4$, to within 0.012. What changes is only how many idle blocks sit in front: ten of sixteen at full depth, two or three of eight. The modality budget is likewise depth-invariant. This is direct mechanistic support for the retrieval result that `layers8` matches `layers12` on `rank@1`.

1 Setup

Two measurements are repeated verbatim from the 16-layer decomposition, on checkpoints from the depth sweep.

Attribution. The exact additive decomposition of the final CLS token into per-modality contributions plus an FFN, bias and initialisation residual. Shares sum to 1 by construction; the reported `budget_sums_to` confirms closure to within 1.5×10^{-8} .

Suppression. Zeroing the cross-attention output of the first k or last k blocks and remeasuring cosine against the true fingerprint.

Table 1: Checkpoints. The two 8-layer arms stopped at different epochs, which is the expected behaviour for `val/mean_cos` with `patience: 30`.

Arm	Checkpoint	Layers	n molecules	Baseline cos
<code>layers16 (final1)</code>	<code>best.ckpt</code>	16	256 / 512	0.9529 / 0.9546
<code>layers8-s0</code>	<code>epoch_epoch=728.ckpt</code>	8	2048	0.9536
<code>layers8-s1</code>	<code>epoch_epoch=631.ckpt</code>	8	2048	0.9533

The 8-layer arms are measured on 2048 molecules, the 16-layer reference on 256 (attribution) and 512 (suppression). The sample sizes differ because the 16-layer numbers were reused from the earlier run rather than recomputed; this is the main caveat and is revisited in §5.

2 Result 1: the live suffix is 5–6 blocks at either depth

Table 2 gives the cost of suppressing the first k blocks. Define the *inert prefix* as the largest k whose suppression costs less than 0.01 cosine, and the *live suffix* as the remaining blocks.

Table 2: Cosine drop from suppressing the first k cross-attention blocks. Dashes mark k values not measured for that arm.

k	1	2	3	4	5	6	8	10	12
layers16	0.0000	0.0000	0.0000	0.0000	−0.0001	0.0002	0.0010	0.0072	0.2523
layers8-s0	−0.0001	0.0017	0.0192	0.1443	0.3452	0.4315	0.7696	—	—
layers8-s1	−0.0002	0.0007	0.0078	0.1507	0.3089	0.4222	0.7565	—	—

- layers16: inert prefix 10, live suffix **6** of 16.
- layers8-s0: inert prefix 2, live suffix **6** of 8.
- layers8-s1: inert prefix 3, live suffix **5** of 8.

The live suffix is the same 5–6 blocks in all three checkpoints. Halving the depth did not halve the amount of computation; it removed idle blocks.

Both depths also show a sharp cliff rather than a gradual decline — at 16 layers between $k = 10$ and $k = 12$, at 8 layers between $k = 3$ and $k = 4$. In both cases the cliff sits about six blocks from the end.

3 Result 2: the final blocks are quantitatively depth-invariant

The stronger version of the claim is in Table 3: it is not merely that a similar *number* of blocks is live, but that they do the same *amount* of work.

Table 3: Cosine drop from suppressing the last k blocks. The 8- and 16-layer columns agree to within 0.012 for every $k \leq 4$, despite a $2\times$ difference in total depth.

k	1	2	3	4	6	8
layers16	0.0368	0.1133	0.1935	0.2720	0.4498	0.6829
layers8-s0	0.0392	0.1125	0.1840	0.2600	0.5402	0.7696
layers8-s1	0.0382	0.1140	0.1898	0.2605	0.5500	0.7565
max spread	0.0024	0.0015	0.0095	0.0120	0.1002	0.0867

For $k \leq 4$ the three checkpoints are interchangeable. The curves separate only at $k \geq 6$, which is exactly where the 8-layer model has run out of blocks and is being asked to give up its inert prefix as well.

Read together with Table 2, the picture is that MARINA’s cross-attention stack does a fixed amount of work in a fixed number of blocks at the end, and pads the front with however many identity-like blocks the architecture provides.

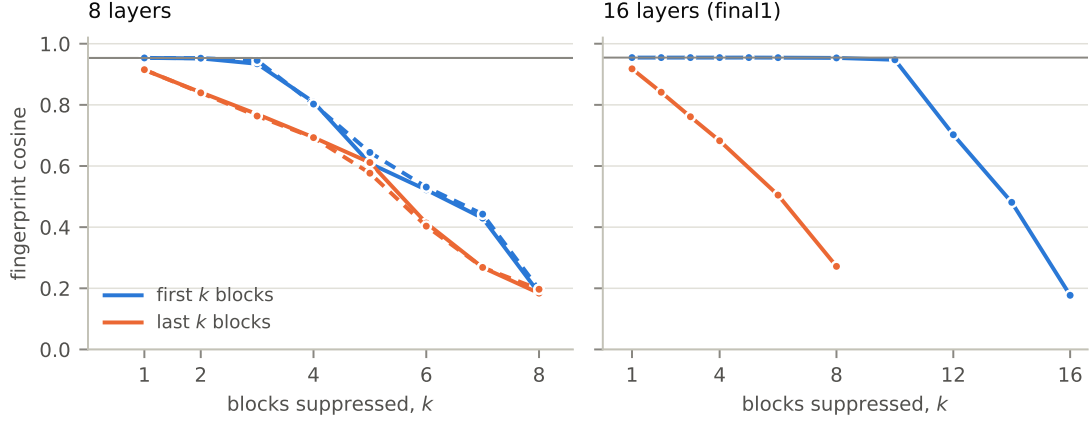


Figure 1: Suppression curves at both depths. The last- k curves overlie one another; the first- k curves are shifted by the difference in inert prefix.

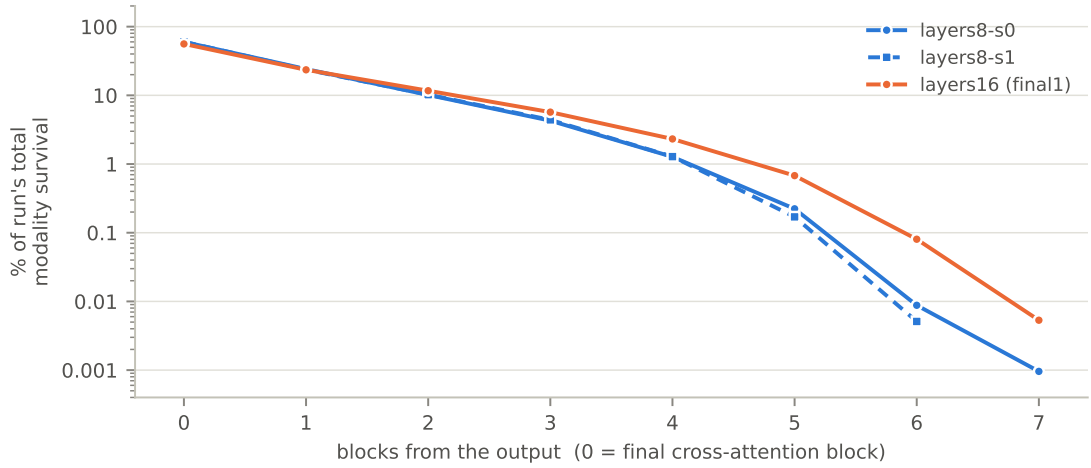


Figure 2: The same curves aligned on distance from the final block, which is the coordinate in which they coincide.

4 Result 3: the modality budget is also depth-invariant

Table 4 gives the peaks-only modality shares. Halving the depth leaves them essentially unchanged, and preserves the ordering $\text{hsqc} \gg \text{c_nmr} > \text{mass_spec} > \text{h_nmr} \gg \text{mw}$.

Table 4: Share of the peaks-only modality budget, and the FFN residual as a fraction of the whole CLS direction.

Arm	hsqc	c_nmr	mass_spec	h_nmr	mw	FFN
layers16	0.4585	0.2438	0.1633	0.1254	0.0090	0.7128
layers8-s0	0.4319	0.2642	0.1810	0.1134	0.0095	0.7031
layers8-s1	0.4274	0.2599	0.1870	0.1148	0.0108	0.7210

The FFN path writes 70–72% of the CLS direction at both depths, and `mw` remains inert at about 1%. The largest single shift is `hsqc`, down about 3 points at 8 layers — comparable to the 0.5-point spread between the two 8-layer seeds, so not clearly a depth effect.

Superadditivity of the joint $\{\text{hsqc}, \text{c_nmr}, \text{h_nmr}\}$ ablation persists: $9.37\times$ (s0) and $8.92\times$ (s1) at 8 layers against $11.09\times$ at 16. Attribution and ablation continue to rank the modalities consistently, with Spearman 0.9 at 8 layers and 0.7 at 16.

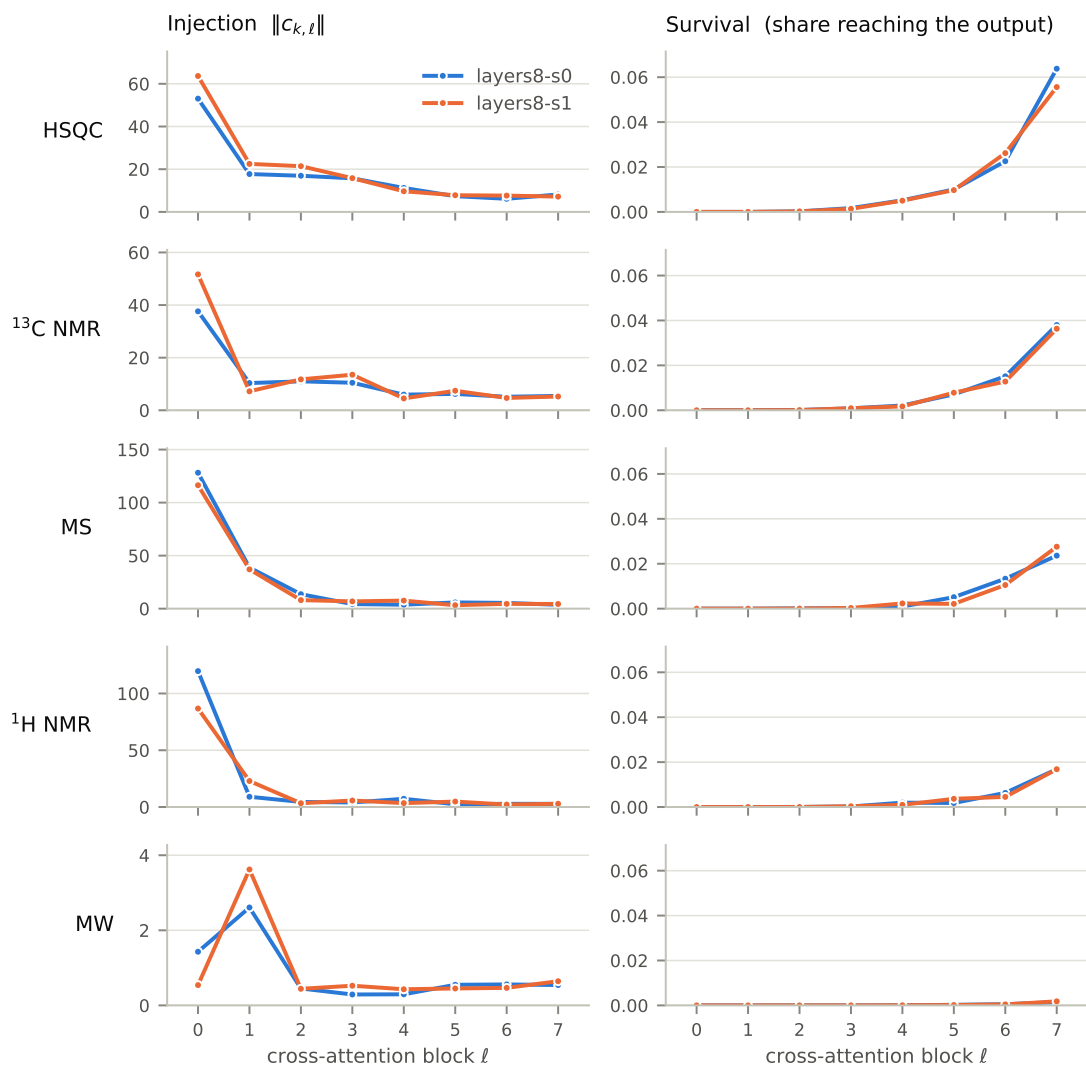


Figure 3: Per-block, per-modality survival and injection profile for the 8-layer arms.

5 Caveats

1. **Unequal sample sizes.** The 8-layer arms use $n = 2048$; the 16-layer reference uses $n = 256$ (attribution) and $n = 512$ (suppression). The 16-layer column is the noisier one, so the agreement in Table 3 is if anything understated — but a matched- n rerun would make the comparison clean, and is cheap.
2. **The 16-layer reference is not the same measurement as the published one.** These numbers come from `final1/best.ckpt` at $n = 256$. The modality-attribution write-up quotes $9.2\times$ superadditivity and Spearman 1.0 from a different run at different n . Both are correct for what they measure; quoting across them is not.
3. **Two seeds, not three.** `layers8-s2` finished after these measurements were taken and is not included.
4. **Suppression is not retraining.** Zeroing a block’s attention output in a trained network measures what that block currently contributes, not what a network trained without it would do. The retrieval sweep is the experiment that answers the latter; this note explains its result.

6 Conclusion

Halving MARINA’s cross-attention depth does not change what the stack computes. The live suffix is 5–6 blocks either way, the cost of suppressing the last k blocks matches across depths to within 0.012 for $k \leq 4$, and the modality budget is unchanged. The eight blocks removed from the 16-layer model were part of an inert prefix that contributed under 0.01 cosine in total.

This is the mechanistic reason `layers8` matches `layers12` on test `rank@1`, and it predicts that further trimming would start to cost something between six and five blocks, where the cliff sits.